Chenkai Niu
Assignment 3 — Decision Making & Control Flow
This assignment has two parts: a short multiple-choice check on Lesson 3 concepts, and two small coding problems that put if / else if / else and Boolean logic to work.
Part 1: Multiple Choice (__ / 8 points)

Check the box next to the best answer for each question. Each question is worth 5 points.

1. In a flowchart, what does a diamond symbol represent? (5 pts) [DAY 1]
☐ A) A start or end point ☐ B) An action step
X C) A decision point ☐ D) A loop

2. Which of the following correctly checks whether x equals 10 (rather than assigning it)? (5 pts) [DAY 2]
☐ A) if (x = 10) **X B) if (x == 10)**
☐ C) if (x := 10) ☐ D) if (x =! 10)

3. What does the && (AND) operator require in order to evaluate to true? (5 pts) [DAY 1]
☐ A) Only one side must be true **X B) Both sides must be true**
☐ C) Neither side can be true ☐ D) Exactly one side must be false

4. Given int x = 45;, what does this code print? (5 pts) [DAY 2]
if (x >= 60) { cout << "Pass"; } else { cout << "Fail"; }

☐ A) Pass **X B) Fail** ☐ C) Nothing prints ☐ D) Compiler error

5. In an else-if chain, C++ checks the conditions: (5 pts) [DAY 2]
☐ A) In random order ☐ B) From bottom to top
X C) Top to bottom, stopping at the first true one ☐ D) All conditions at the same time

6. Which value is treated as false in C++? (5 pts) [DAY 1]
☐ A) Any negative number **X B) Only the number 0**
☐ C) Any decimal number ☐ D) The word “false” only

7. What is the value of (7 > 3) || (2 == 5)? (5 pts) [DAY 1]
X A) true ☐ B) false ☐ C) Compiler error ☐ D) Undefined

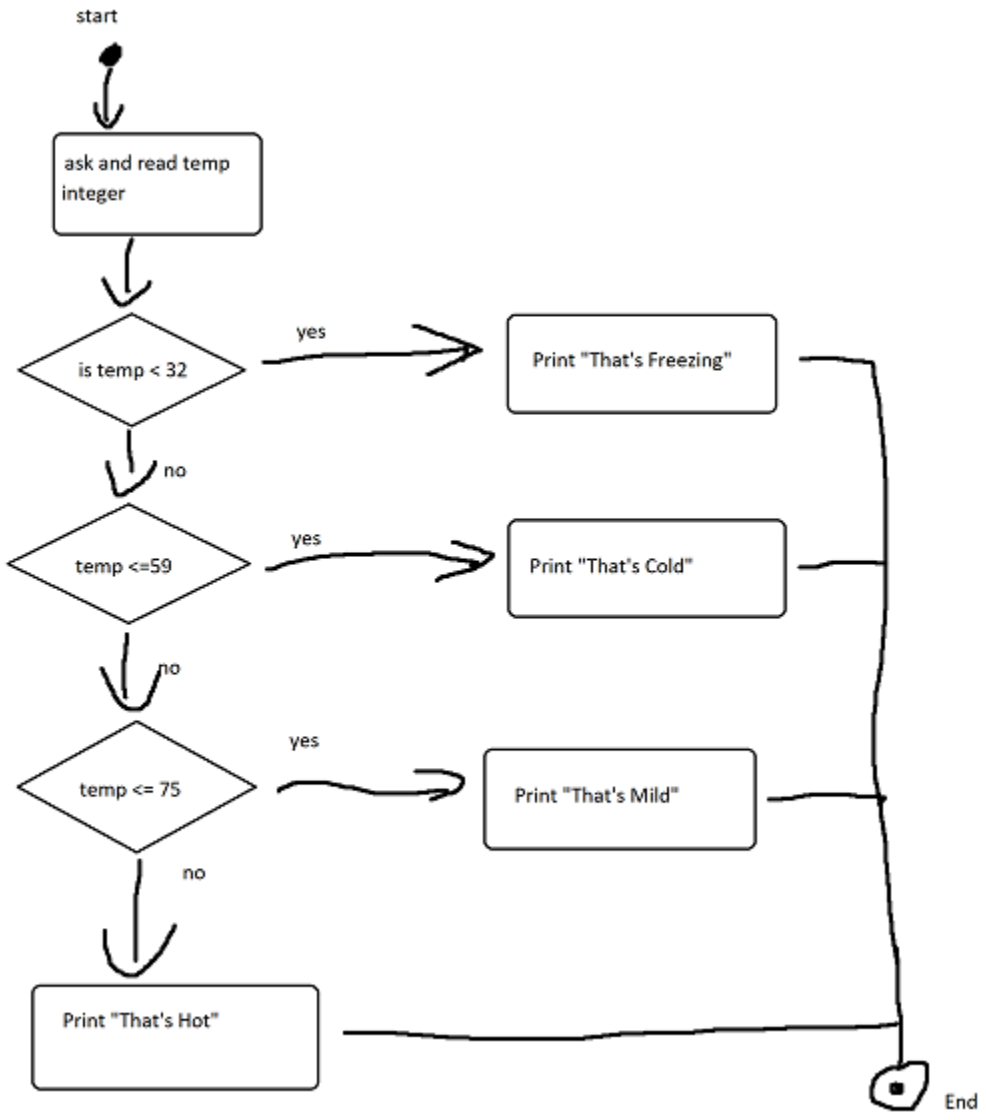
8. What happens if you accidentally write if (grade = 60) instead of if (grade == 60)? (5 pts) [DAY 2]
☐ A) The compiler catches the error immediately
X B) It assigns 60 to grade and the condition always evaluates as true
☐ C) It behaves identically to ==
☐ D) The program refuses to compile

Part 2: Coding Problems (__ / 22 points)

Since we won't be meeting Monday, these three problems cover the hands-on practice you would have gotten in that session — Boolean logic, relational operators, and if/else-if chains, each from a slightly different angle. Each problem is worth 20 points.

Problem 1: Temperature Classifier (__ / 15 pts) [DAY 2]

(__ / 5 points) Draw the flow chart



(__ / 10 points) Write a C++ program called temp_check.cpp that classifies a Fahrenheit temperature.

☐ Ask the user to enter a temperature (as an integer) using cin.
☐ Use an if / else if / else chain to classify it using these rules:
Below 32 → “Freezing” 32–59 → “Cold” 60–75 → “Mild” 76 and above → “Hot”
☐ Print the classification clearly, e.g. “That's Mild.”

Example interaction:
Enter the temperature (F): 70
That's Mild.

```
// Name: Chenkai Niu
// Course: CSCI 271 01
// Assignment 3 - Problem 1

#include <iostream> // I/O library for input and output operations
using namespace std; // Dont need to std:: every line

int main() { // Main function, start point
    int temp; // makes an integer variable to hold the user's input
temperature
    cout << "Enter the temperature (F): "; // Prints instructions/asks the
user to enter a temperature in Fahrenheit
    cin >> temp; // reads user input and stores in temp
    if (temp < 32) { // checks if the temperature is strictly less than 32
        cout << "That's Freezing." << endl; // if it is, prints
    } else if (temp <= 59) { // checks if temperature is between 32 and 59
        cout << "That's Cold." << endl; // if it is, prints
    } else if (temp <= 75) { // checks if temperature is between 60 and 75
        cout << "That's Mild." << endl; // if it is, prints
    } else { // Handles any temperature that is 76 or above
        cout << "That's Hot." << endl; // prints for all temps
    }
    return 0; // return 0 is the end
}
```

Problem 2: Discount Eligibility (__ / 7 pts) [DAY 1 & 2]

Write a C++ program called eligibility.cpp that checks whether someone qualifies for a discount.

☐ Ask the user for their age (an integer).
☐ Ask whether they're a member, entered as 1 for yes or 0 for no.
☐ A person qualifies if their age is 60 or older, OR if they're 18+ AND a member.

☐ This means combining && and || in one condition — use parentheses to make your intent clear, e.g. (age >= 18 && isMember == 1).
☐ Print whether they qualify or not, clearly.

Example interaction:
Enter your age: 20
Are you a member? (1 = yes, 0 = no): 1
You qualify for the discount.

```
// Name: Chenkai Niu
// Course: CSCI 271 01
// Assignment 3 - Problem 2

#include <iostream> // I/O library for input and output operations
using namespace std; // Dont need to std:: every line

int main() { // Main function, start point
    int age; // makes an integer variable to store the user's age
    int isMember; // makes an integer variable to store membership status
(1 = yes, 0 = no)
    cout << "Enter your age: "; // prints question/command for the user to
enter their age
    cin >> age; // reads the age input from the user and stores into the
age variable
    cout << "Are you a member? (1 = yes, 0 = no): "; // asks user for
their membership status
    cin >> isMember; // reads the membership response and stores into the
isMember variable
    if (age >= 60 || (age >= 18 && isMember == 1)) { // checks if age is
60+ OR || (18+ AND && a member, needs == to check for input of 1)
        cout << "You qualify for the discount." << endl; // if either
pass, print
    } else { // if the user does not meet either qualification condition
        cout << "You do not qualify for the discount." << endl; // prints
rejection message
    }
    return 0; // return 0 is end
}
```

Before You Submit

- ☐ All three programs compile and run without errors on your machine.
- ☐ Files are named temp_check.cpp, even_odd.cpp, and eligibility.cpp.
- ☐ A header comment at the top of each file with your name and “Assignment 3.”
- ☐ Each program tested with at least two different inputs, covering different branches.
- ☐ Comment out every line of the code